

Unit-I
Numerical techniques
Numerical solution of algebraic and transcendental
equations:

Newton - Raphson methods



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Newton-Raphson Method

This is also an **iteration method** and is used to find the roots of an equation $f(x) = 0$, when the **derivative** of $f(x)$ is a **simple expression**.

It is derived as follow:

Let $x = x_0$ be an **approximate value** of one root of the equation $f(x) = 0$ in $[a, b]$. If $x = x_1$ is the exact root then

$$f(x_1) = 0, \quad (1)$$

where the difference between x_0 and x_1 is very small and if h denote the difference then

$$x_1 = x_0 + h. \quad (2)$$

Now from equation **(1)**, we have

$$f(x_0 + h) = 0$$

By Taylor's theorem, we get

$$f(x_0) + \frac{h}{1!} f'(x_0) + \frac{h^2}{2!} f''(x_0) + \frac{h^3}{3!} f'''(x_0) + \dots = 0.$$

Since h is small, **neglecting the terms** containing $h^2, h^3, \dots$, we get

$$f(x_0) + \frac{h}{1!} f'(x_0) = 0 \Rightarrow h = -\frac{f(x_0)}{f'(x_0)}.$$

Now from equation **(2)**, we get

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

The above value of x_1 is a **closer approximation** to the root of $f(x) = 0$ than x_0 . Similarly if x_2 denotes a **better approximation**, starting with x_1 , then

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}.$$

Proceeding in this way, we get

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \quad (3)$$

Equation **(3)** is Newton-Raphson formula.

Example: Find one root of $x^4 - x - 10 = 0$ between 1 and 2 by Newton-Raphson method Answer should be correct to 3 decimal places.

Solution: Here $f(x) = x^4 - x - 10 \Rightarrow f'(x) = 4x^3 - 1$
and

$$f(1) = -10 = -\text{ve}, f(2) = 4 = +\text{ve}.$$

Since 4 is close to 0, so take $x_0 = 2$ as initial approximation.

Hence **first approximation** is

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$
$$\Rightarrow x_1 = 2 - \frac{2^4 - 2 - 10}{4 \cdot 2^3 - 1} = 1.871.$$

Second approximation is

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\Rightarrow x_2 = 1.871 - \frac{(1.871)^4 - 1.871 - 10}{4.(1.871)^3 - 1} = 1.856.$$

Third approximation is

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$\Rightarrow x_3 = 1.856 - \frac{(1.856)^4 - 1.856 - 10}{4.(1.856)^3 - 1} = 1.856.$$

Since $x_3 = x_2$. Hence the desired root is 1.856 (correct to three decimal places).

Q.1: Find one root of $3x - \cos x - 1 = 0$ between 0 and 1 by Newton-Raphson method. Answer should be correct to 4 decimal places.

Answer: 0.6071.

Q.2: Find a root of the equation $x^3 - 2x - 5 = 0$ by Newton-Raphson method. Answer should be correct to three decimal places.

Answer: 2.094.

Thank You